

Postmagmatic Alteration Processes — Complete Concept Guide

Autometamorphism · Deuteric Alteration · Devitrification · Replacement Textures

References: Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed. & Perkins, *Mineralogy*, 3rd ed.

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1. What Counts as "Alteration"? Autometamorphism vs. Deuteric Alteration

Winter groups all of this under **secondary textures**: changes that happen to an igneous rock's minerals *after* the rock is essentially solid, but while the surrounding pluton is still hot from its own heat of crystallization — not from any later, separate metamorphic event.

"Secondary mineral reactions that occur in igneous rocks as they cool, and that are not products of a distinct later metamorphic event, are commonly called autometamorphic processes rather than metamorphic ones because they are a natural part of igneous cooling. Autometamorphic processes are more common in plutonic rocks than in volcanics because they remain at elevated temperatures for a longer time."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Important boundary case

Winter explicitly notes: "Diagenetic and weathering processes are not considered to be autometamorphic (a rather arbitrary distinction)." So if an exam option offers "diagenesis" or "weathering" as an alternative to a named deuteric process, it is always wrong — those are excluded by definition, even though the line is admittedly a judgment call.

Deuteric alteration: The specific subset of autometamorphic reactions that involve **hydration** — water either escaping from the last residual melt, or entering from outside, reacting with already-crystallized minerals at moderate temperature.

"Most, but not all, autometamorphic reactions involve minerals at moderate temperatures in an environment in which H₂O is either liberated from residual melt or externally introduced. Such alterations are a subset of autometamorphism, which involves hydration, and they are called deuteric alterations."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Note the general pattern that runs through nearly every deuteric reaction below: an anhydrous, higher-temperature igneous mineral is progressively replaced by a lower-temperature, **hydrous** mineral, and replacement typically starts at grain rims/cleavages and works inward.

"As in other deuteric alterations, hydration attacks the outer margin of a mineral, so that chlorite generally replaces the initial mafic from the rim inward. In the case of biotite, H₂O may work its way along the prominent cleavages."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

2. The Deuteric (Hydration) Alteration Reactions

(a) Uralitization — Pyroxene → Amphibole

"Pyroxene is a common primary mafic mineral in a variety of igneous rocks. If H₂O penetrates at modest temperatures, a deuteric alteration of pyroxene to amphibole results, called uralitization. Any gradation from amphibole rims on pyroxene cores to multiple patches of pyroxene in an amphibole to complete replacement is possible. The amphibole may be a single crystal of hornblende or a fibrous actinolite or hornblende aggregate. Either, when demonstrated to result from pyroxene alteration, may be called uralite, but the term is more commonly applied to the aggregates."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Table 1 shorthand: "Replacement of pyroxene by amphibole." (p. 55)

(b) Biotitization — Pyroxene/Amphibole → Biotite

"Biotitization is a similar process of hydration/deuteric alteration that produces biotite, either directly from pyroxene, or, more commonly, from hornblende. Because biotite contains little Ca, epidote may be produced as Ca is released during the alteration of hornblende to biotite."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Table 1 shorthand: "Replacement of pyroxene or amphibole by biotite." (p. 55)

Exam trap

Biotitization is another reaction that can release Ca to form epidote as a by-product — don't confuse this incidental epidote with the epidote that defines saussuritization; the diagnostic reactant is different (hornblende here, plagioclase in saussuritization).

(c) Chloritization — Any Mafic Mineral → Chlorite

"Chloritization is the alteration of any mafic mineral to chlorite. Chlorite is a very hydrous phyllosilicate and typically replaces the less hydrous mafics at low temperature when water is available. Pyroxenes, hornblendes, and biotites are commonly observed in thin section in various stages of alteration to chlorite."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Table 1 shorthand: "Replacement of any mafic mineral by chlorite." (p. 55)

(d) Sericitization — Feldspar/Feldspathoid → White Mica

"Sericitite is a term applied to any very fine-grained white mica. Seritization is the process by which felsic minerals (usually feldspars or feldspathoids in igneous rocks) are hydrated to produce sericite. Incipient stages can be recognized by a fine, dusty appearance of feldspars in plane-polarized light... K⁺ ions are required for plagioclase to be altered to the common forms of sericite. Potassium may be released by chloritization of nearby biotite. K-feldspar requires no additional K⁺ and may be more sericitized than associated plagioclase."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

Table 1 shorthand: "Replacement of feldspar or feldspathoids by fine white micas." (p. 55)

(e) Saussuritization — Ca-Rich Plagioclase → Epidote + Albite

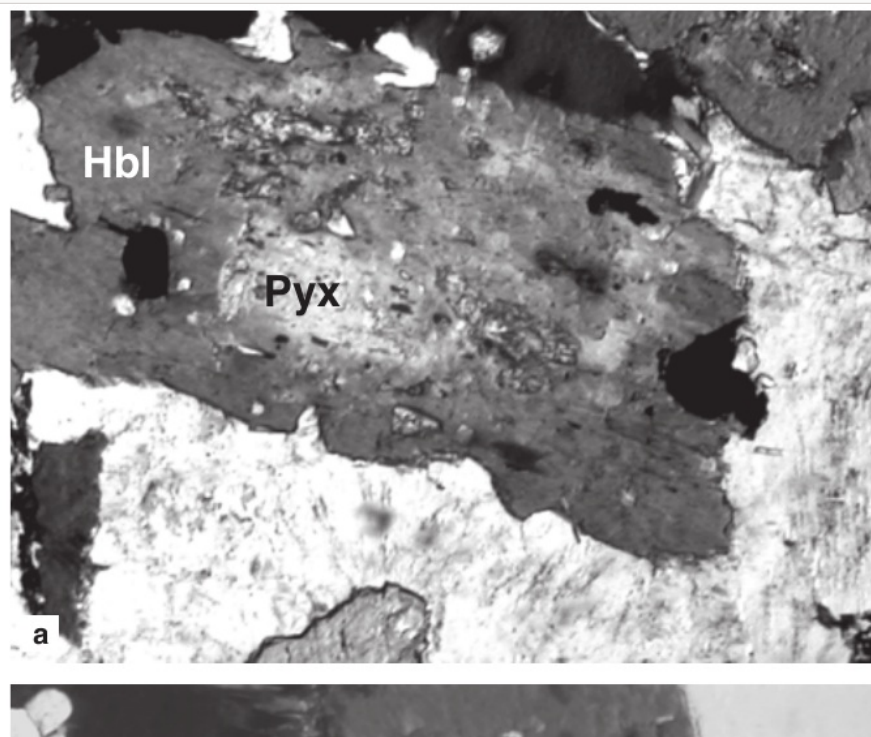
"Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more nearly pure albite, releasing Ca and Al to form an epidote mineral ($\pm$ calcite and/or sericite). In zoned plagioclase, one can occasionally observe the products of saussuritization or seritization concentrated in the Ca-rich core or some of the more Ca-rich oscillatory bands in the original grain."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 51-52

Table 1 shorthand: "Replacement of plagioclase by epidote." (p. 55)

"Sausserite is a name given to fine-grained epidote produced by alteration of plagioclase."

— Perkins, *Mineralogy*, p. 391



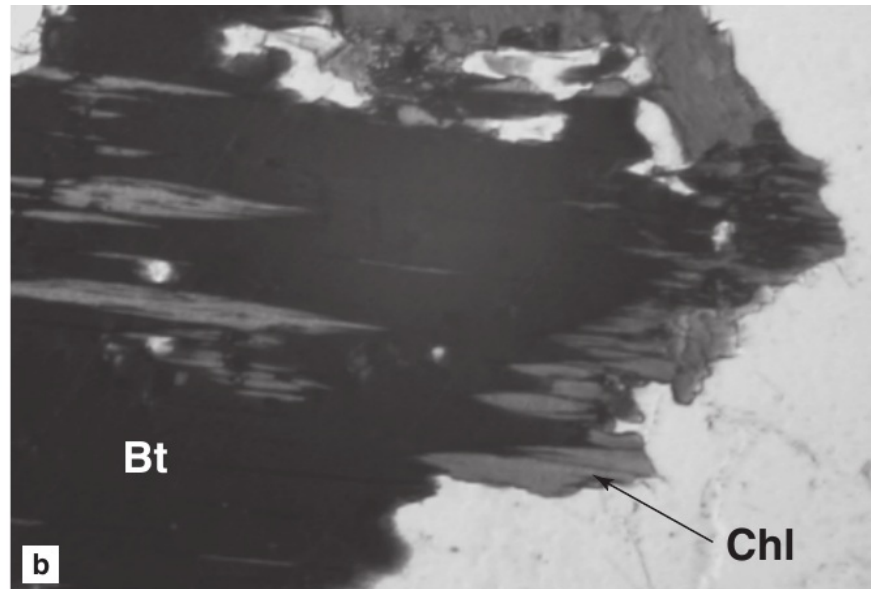


FIGURE 20 (a) Pyroxene largely replaced by hornblende. Some pyroxene remains as light areas (Pyx) in the hornblende core. Width 1 mm. (b) Chlorite (light) replaces biotite (dark) at the rim and along cleavages. Tonalite, San Diego, California. Width 0.3 mm.

generally replaces the initial mafic from the rim inward. In the case of biotite, H_2O may work its way along the prominent cleavages, and chlorite can be seen to replace biotite margins as well as along cleavage planes (Figure 20b).

Sericite is a term applied to any very fine-grained white mica. **Seritization** is the process by which felsic minerals (usually feldspars or feldspathoids in igneous rocks) are hydrated to produce sericite. Incipient stages can be recognized by a fine, dusty appearance of feldspars in plane-polarized light. In more advanced stages of alteration, the feldspars appear speckled, with fine micas having yellowish birefringence and then larger clots with coarser crystals and higher birefringence. K^+ ions are required for plagioclase to be altered to the common forms of sericite. Potassium may be released by chloritization of nearby biotite. K-feldspar requires no additional K^+ and may be more sericitized than associated plagioclase.

Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more nearly pure albite,

Figure 20 (Winter, Principles of Igneous and Metamorphic Petrology, p. 53) — (a) Pyroxene largely replaced by hornblende, remnant pyroxene as light patches (uralitization). (b) Chlorite (light) replacing biotite (dark) at rims and along cleavages (chloritization), tonalite, San Diego.

3. Olivine Alteration — Serpentine & Iddingsite

"Olivine is easily altered in cooling mafic rocks, even in volcanics. It is commonly rimmed or replaced by serpentine or dark brown iddingsite."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

Note that olivine's alteration is distinctive enough to have its own dedicated pair of product minerals (serpentine, a hydrous Mg-phyllosilicate; iddingsite, a reddish-brown mixture of iron oxides/oxyhydroxides and clay), rather than falling under one of the five named "-ization" processes above.

4. Symplectite & Pseudomorphs — the General Replacement Concept

Pseudomorph: A replacement texture in which one or more minerals replace another, but the crystal retains the outward form of the original mineral.

Symplectite: A replacement texture in which a mineral is replaced by an intergrowth of two or more minerals growing together.

"Symplectite is a term applied to fine-grained intergrowths resulting from the combined growth of two or more minerals as they replace another mineral. As in any of the replacements described above, replacement may be partial or complete. Complete replacement and pseudomorphs are common. The fibrous actinolite-hornblende 'uralite' aggregates replacing pyroxene are an example. Biotite and epidote replacing hornblende is another."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

How this connects

Symplectite is the general textural category; uralite and biotite+epidote (from biotitization) are two named examples of symplectites that already appeared above. An exam question describing "an intergrowth of two minerals replacing an original grain" without naming a specific reaction is testing the term *symplectite*, not a specific "-ization."

5. Myrmekite — A Related (Non-Hydration) Secondary Texture

"Myrmekite is an intergrowth of dendritic quartz in a single crystal of plagioclase. The quartz appears rod-like in thin section, and numerous adjacent rods go extinct in unison, indicating that they are all parts of a single quartz crystal. Myrmekites are very common in granitic rocks and occur preferentially where plagioclase is in contact with K-feldspar. Myrmekites appear to have grown from the plagioclase-K-feldspar boundary into the K-feldspar. As the plagioclase replaces the K-feldspar, SiO₂ is released (the anorthite component of plagioclase contains less SiO₂ than the K-feldspar), thereby producing the quartz."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 52-53

Key distinction for MCQs

Myrmekite is **not** a hydration/deuteric reaction — it's driven by a silica-releasing plagioclase-K-feldspar replacement, not by water attacking a mafic or feldspathic mineral. If an option lists "myrmekite" alongside saussuritization/uralitization/chloritization as if they're all the same kind of process, that's the distractor to catch.

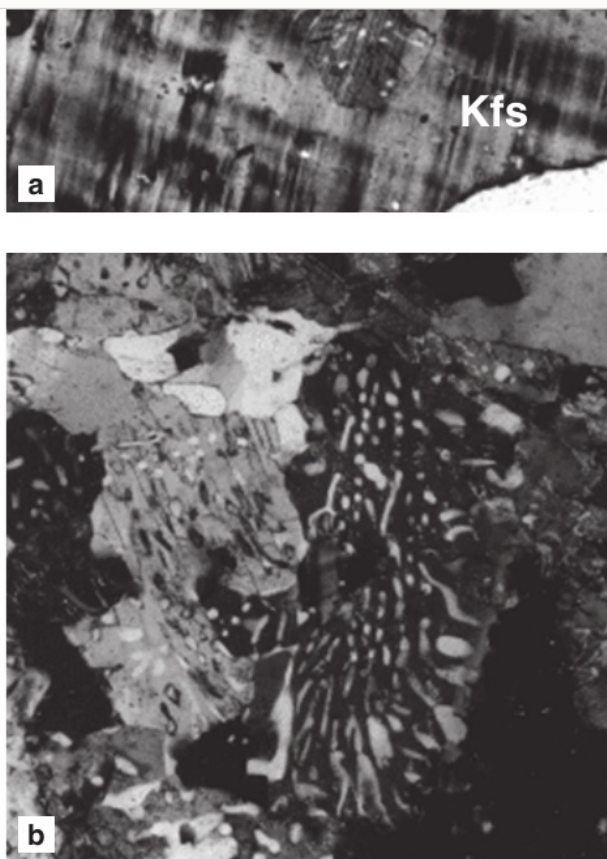


FIGURE 21 Myrmekite formed in plagioclase at the boundary with K-feldspar. Photographs courtesy L. Collins.

Figure 21 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 53) — Myrmekite formed in plagioclase at the boundary with K-feldspar (Plag, Kfs labeled). Note the wormy, dendritic quartz rods.

6. Devitrification of Glass

"Devitrification is the secondary crystallization of glass to fine-grained mineral aggregates. Glass is an inherently unstable material and is readily replaced by more stable minerals when kinetics permits. Water-quenched basaltic glass may be replaced by brown, optically isotropic oxidation-hydration products collectively known as palagonite. More silicic glassy rocks commonly devitrify to produce a microgranular mass of small, equidimensional grains of interlocking feldspar and silica minerals called felsitic texture. Crystal form is entirely suppressed in felsitic rocks, and the texture looks very much like that of chert. Devitrification of glass may also produce radial aggregates of crystals (commonly cristobalite or tridymite plus feldspar) called spherulites. Lithophysae are large cavities bordered by spherulitic growth... The minerals produced by devitrification are generally too fine to identify under the polarizing microscope, and x-ray analysis may be required."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 52-53

Devitrification Product	Description
Palagonite	Brown, optically isotropic oxidation-hydration product of water-quenched basaltic glass
Felsitic texture	Microgranular interlocking feldspar + silica minerals from silicic glass; resembles chert
Spherulites	Radial aggregates of cristobalite/tridymite + feldspar
Lithophysae	Large cavities bordered by spherulitic growth, common in rhyolites (late gas bubbles)

Why devitrification is always the wrong answer for "plagioclase → albite + epidote" questions

Devitrification only ever acts on **glass** (an amorphous, non-crystalline material). It cannot be the answer to any question describing the alteration of an already-crystalline mineral like plagioclase, pyroxene, or biotite — that's an automatic elimination clue.

7. Quick-Reference Summary Table

Process	Starting Material	Product	Category
Uralitization	Pyroxene	Amphibole (uralite)	Deuteric
Biotitization	Pyroxene / amphibole	Biotite ($\pm$ epidote as Ca by-product)	Deuteric
Chloritization	Any mafic mineral	Chlorite	Deuteric
Sericitization	Feldspar / feldspathoid	Sericite (fine white mica)	Deuteric
Saussuritization	Ca-rich plagioclase	Albite + epidote ($\pm$ calcite, sericite)	Deuteric
Olivine alteration	Olivine	Serpentine or iddingsite	Deuteric
Myrmekite formation	Plagioclase + K-feldspar boundary	Dendritic quartz + plagioclase	Non-hydration secondary texture
Devitrification	Volcanic glass	Palagonite / felsite / spherulites / lithophysae	Secondary crystallization (not deuteric hydration of a crystal)

8. Probable Exam Questions

Q. Which deuteric process converts pyroxene or amphibole to biotite, sometimes releasing Ca to also form epidote?

A. Biotitization.

Q. What potassium source allows plagioclase (rather than K-feldspar) to sericitize?

A. K⁺ released by nearby chloritization of biotite; K-feldspar itself needs no additional K⁺ and may sericitize more readily.

Q. What are the two characteristic alteration products of olivine?

A. Serpentine and dark brown iddingsite.

Q. What textural term describes a mineral being replaced by an intergrowth of two or more minerals?

A. Symplectite.

Q. Where does myrmekite preferentially form, and what drives its growth?

A. At plagioclase-K-feldspar boundaries in granitic rocks; growth is driven by excess SiO₂ released as plagioclase replaces K-feldspar (anorthite has less SiO₂ than K-feldspar).

Q. What distinguishes palagonite formation from felsitic devitrification?

A. Palagonite forms from water-quenched basaltic (mafic) glass; felsitic texture forms from more silicic glass crystallizing into interlocking feldspar + silica minerals.

Q. Are diagenesis and weathering considered autometamorphic processes?

A. No — Winter explicitly excludes both, even though he calls the distinction "rather arbitrary."